GRISHMA BALGI

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Embedded Systems & Firmware

EDUCATION

New York University | May 2027
M.S. Mechatronics and Robotics | GPA: 3.834/4.0

Veermata Jijabai Technological Institute | June 2025
B.Tech. Mechanical Engineering; Honors in Automobile Engineering | GPA: 9.04/10

TECHNICAL SKILLS

Languages & Tools: C, C++, Python, ARM Thumb-2 assembly, Mbed OS

Embedded Systems: STM32L475, Arduino, Parallax Propeller, interrupts, timers, ring buffers

Interfaces & Algorithms: I2C, UART/serial, HX711 acquisition, filtering, dynamic time warping

EXPERIENCE

Prema Lab, New York University | Sep. 2025 - Present
Graduate Research Assistant | Robotics fabrication, instrumentation and simulation

- Fabricated a robotic-fin test rig and automated Silhouette Portrait 4 cutting through Python USB commands, generating coordinate-based paths for candidate fin geometries.
- Developed a motor-control interface and Arduino acquisition firmware to vary oscillation waveform, amplitude and frequency; logged load-cell readings alongside commanded and measured motor positions.

Schneider Electric | June 2024 - June 2025
Engineering Intern | Mechanical modelling and sensor systems

- Worked on reduced-order modelling of mechanical systems using Creo and Altair tools, preparing simulation-derived data and exploring models for circuit-breaker mechanisms.
- Built experience in sensor selection, vibration data acquisition and Python-based analysis for circuit-breaker condition monitoring.

SELECTED PROJECTS

KinetiKey | Gesture Combination Lock
C++, Mbed OS, STM32L475, I2C, DTW | Course project

- Implemented IMU-based gesture authentication on STM32L475 using I2C sensor acquisition, low-pass filtering and band-constrained dynamic time warping.
- Used static buffers for three-gesture sequences of up to 200 samples per gesture; added interrupt-driven button input, enrollment logic, failed-attempt lockout and LED state feedback.

ECG Acquisition and Heart-Rate Monitor
C++, ARM Thumb-2 assembly, STM32, Mbed | Course project

- Implemented periodic ECG acquisition with an Mbed Ticker, a 1,080-sample circular buffer and digital lead-off detection on an STM32 platform.
- Wrote an ARM assembly peak-counting routine with an amplitude threshold, refractory interval and heart-rate range checks, alongside a C reference implementation.

FlexiClimber 2.0 | Beam-Traversal Robot
C, Arduino, Parallax Propeller, SolidWorks | Course project

- Extended an existing rack-and-pinion beam-traversal platform with a pivot mechanism; integrated mechanical components, actuators and command handling for reorientation.
- Worked with Propeller C firmware for concurrent sensing, Bluetooth commands and actuator sequencing, alongside Arduino pivot-control routines; full pivot integration remained unfinished.